A CASE-STUDY BASED REPORT

ON

PRODUCT REVIEW SENTIMENT ANALYSIS USING NLP AND BERT

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ABSTRACT

Product reviews provide valuable information about customers' opinions, experiences, and satisfaction with products. However, manually analyzing a large number of reviews is time-consuming and difficult. Sentiment analysis using Natural Language Processing (NLP) provides an automated approach for identifying whether a review expresses a positive or negative opinion.

This project presents a Product Review Sentiment Analysis system using NLP and BERT. The system uses an Amazon product review dataset containing 4,915 reviews with information such as reviewer details, review text, ratings, review summaries, and helpfulness information. Exploratory Data Analysis (EDA) was performed to understand the structure, missing values, rating distribution, review length, word frequencies, and other characteristics of the dataset.

For sentiment classification, product ratings were converted into binary sentiment labels. Ratings of 4 and 5 were classified as Positive, while ratings of 1, 2, and 3 were classified as Negative. After removing one review with missing text, 4,914 reviews were used for model development.

Two approaches were implemented. First, a traditional NLP baseline using TF-IDF with Logistic Regression was developed. Second, a pretrained BERT (Bidirectional Encoder Representations from Transformers) model was fine-tuned for sentiment classification. The TF-IDF and Logistic Regression model achieved an accuracy of 91.86%, while BERT achieved an accuracy of 94.51%. BERT also achieved an F1-score of 97.03%, demonstrating better overall classification performance.

The final system can also accept a new product review and predict its sentiment along with a confidence score. The project demonstrates how traditional NLP techniques and transformer-based models can be applied to automatically analyze customer opinions.

INTRODUCTION

1.1 Background

The rapid growth of e-commerce and online shopping has resulted in a significant increase in the amount of customer-generated data available on digital platforms. Among the different types of customer-generated information, product reviews are one of the most valuable sources because they contain direct feedback about customers' experiences, opinions, satisfaction levels, and perceptions of products. Customers commonly provide written reviews after purchasing a product, describing aspects such as product quality, usability, performance, reliability, price, packaging, delivery experience, and overall satisfaction. In addition to written comments, many online shopping platforms allow customers to provide numerical ratings, which provide another indication of their satisfaction. As the number of online purchases continues to increase, the number of product reviews also grows rapidly, making these reviews an important source of information for both customers and businesses.

Product reviews can influence purchasing decisions because potential customers often examine the experiences of previous buyers before purchasing a product. A large number of positive reviews can increase confidence in a product, whereas negative reviews can indicate possible problems or disadvantages. From a business perspective, product reviews provide useful information for understanding customer expectations and identifying areas where products or services can be improved. For example, repeated negative comments about product quality may indicate a manufacturing or design problem, while repeated positive comments may indicate features that customers value. Therefore, analyzing customer reviews can help organizations understand customer satisfaction and make better decisions based on customer feedback.

However, manually analyzing a large collection of product reviews is difficult, time-consuming, and inefficient. When a product receives hundreds or thousands of reviews, it is not practical for a person to read every review individually and determine the overall opinion of customers. Furthermore, different customers may express similar opinions using completely different words and sentence structures. Some reviews may be very short, while others may contain detailed explanations. Reviews can also contain informal language, spelling variations, repeated words, and contextual expressions. These characteristics make automated processing

of review text a challenging task. Therefore, an automated approach is required to process large volumes of textual reviews and identify the sentiment expressed by customers.

Natural Language Processing, commonly known as NLP, provides techniques that allow computers to process, analyze, and interpret human language. NLP is widely used for applications such as text classification, machine translation, information retrieval, question answering, speech processing, and sentiment analysis. In the context of product reviews, NLP can be used to transform unstructured review text into a representation that can be processed by machine learning algorithms. Traditional NLP techniques generally involve text preprocessing and feature extraction methods that convert words or phrases into numerical representations. These numerical representations can then be provided to machine learning algorithms for classification.

Sentiment analysis is an important application of NLP that focuses on identifying the emotional or opinion-based orientation of textual data. In product review analysis, sentiment analysis can determine whether a customer has expressed a positive or negative opinion about a product. For example, a review such as “The product is excellent and works perfectly” indicates a positive sentiment, whereas a review such as “The product stopped working after a few days” indicates a negative sentiment. By automatically classifying reviews according to their sentiment, large collections of customer feedback can be analyzed more efficiently. Sentiment analysis can therefore help businesses obtain an overall understanding of customer opinions without manually examining every individual review.

In this project, product ratings are used to create sentiment labels for the written reviews. Reviews with ratings of four and five are considered positive, while reviews with ratings of one, two, and three are considered negative. This converts the original rating-based problem into a binary sentiment classification problem. The written review text is used as the input to the classification models, while the sentiment label generated from the rating is used as the target variable. This approach allows machine learning models to learn the relationship between the language used in a review and the sentiment associated with that review.

One of the traditional approaches used in this project is Term Frequency–Inverse Document Frequency, commonly known as TF-IDF. TF-IDF is a text representation technique that

converts textual documents into numerical feature vectors based on the importance of words within the documents. The method considers how frequently a term appears in a particular document and how frequently that term appears across the complete collection of documents. Words that occur frequently in a particular review but are less common across the complete dataset can receive higher importance. TF-IDF can therefore provide useful numerical features for traditional machine learning algorithms. In this project, TF-IDF is combined with Logistic Regression to create a traditional NLP baseline model for sentiment classification.

Although traditional NLP techniques such as TF-IDF can provide good results, they have limitations when dealing with the context and meaning of natural language. A word can have different meanings depending on the words surrounding it, and the meaning of a sentence can sometimes depend on word order and context. For example, the difference between “the product is good” and “the product is not good” is strongly influenced by the word “not.” Traditional word-frequency representations may not fully capture such contextual relationships. These limitations have encouraged the development of more advanced language models based on deep learning and transformer architectures.

BERT, which stands for Bidirectional Encoder Representations from Transformers, is a transformer-based language model designed to understand the contextual relationships between words in natural language. Unlike approaches that primarily represent words based on their frequency, BERT processes words in relation to their surrounding context. The bidirectional nature of BERT allows information from both directions of a sentence to contribute to the representation of a word. This contextual understanding makes BERT suitable for language-related tasks such as sentiment classification, text classification, question answering, and other natural language understanding applications.

The proposed project therefore combines traditional NLP and modern transformer-based NLP approaches for product review sentiment analysis. A TF-IDF and Logistic Regression model is first developed as a baseline so that the performance of a traditional machine learning approach can be measured. A pretrained BERT model, specifically the bert-base-uncased model, is then fine-tuned for binary sentiment classification. Comparing these two approaches provides an understanding of how a traditional text representation technique performs against a contextual transformer-based model on the same product review dataset.

The dataset used in this project contains Amazon product reviews along with associated information such as reviewer details, product identifiers, review text, product ratings, review summaries, helpful votes, and review dates. The dataset initially contains 4,915 records and 12 attributes. Exploratory Data Analysis is performed before model development to understand the structure and characteristics of the data. The analysis includes examining the dataset dimensions, data types, missing values, rating distribution, review length, helpful votes, review years, frequent words, and relationships between numerical variables. This analysis provides an understanding of the data and helps identify issues that need to be addressed before training the sentiment classification models.

The dataset contains a strong imbalance between positive and negative reviews. A large proportion of the reviews have high ratings, particularly five-star ratings. After converting ratings into sentiment categories, ratings of four and five are treated as positive, while ratings of one, two, and three are treated as negative. One record containing missing review text is removed because review text is required as the input for sentiment classification. The resulting dataset contains 4,914 usable reviews, consisting of 4,448 positive reviews and 466 negative reviews. The class imbalance is an important characteristic of the dataset because a model can achieve high overall accuracy by performing very well on the majority class while performing less effectively on the minority class.

The dataset is divided into training and testing portions using an 80 percent training and 20 percent testing split. Stratified splitting is used so that the distribution of sentiment classes is maintained between the training and testing datasets. The training dataset contains 3,931 reviews, while the testing dataset contains 983 reviews. The training data is used to learn the relationship between review text and sentiment, while the testing data is kept separate for evaluating how well the trained models perform on previously unseen reviews.

The traditional TF-IDF and Logistic Regression model provides a baseline for comparison. In this implementation, TF-IDF is configured with a maximum of 5,000 features and considers both individual words and two-word combinations. Logistic Regression is then trained using the resulting numerical representation of the review text. This approach provides an efficient and interpretable baseline for text classification and establishes a performance reference against which the BERT model can be evaluated.

The BERT-based approach uses the pretrained bert-base-uncased model and a corresponding BERT tokenizer. The review text is converted into tokens and prepared for the model using a maximum sequence length of 128 tokens. Reviews longer than the specified maximum length are truncated, while shorter reviews are padded to maintain consistent input dimensions. The pretrained BERT model is then fine-tuned for two-class sentiment classification, where the two output classes represent negative and positive sentiment. The model is trained using the prepared training dataset and evaluated using the testing dataset.

The performance of the two models is evaluated using accuracy, precision, recall, and F1-score. These metrics provide different perspectives on classification performance. Accuracy measures the proportion of correctly classified reviews, while precision measures the correctness of positive predictions. Recall measures the ability of the model to identify relevant positive examples, and the F1-score provides a combined measure of precision and recall. Because the dataset contains significantly more positive reviews than negative reviews, considering multiple evaluation metrics is important rather than relying only on accuracy.

The TF-IDF and Logistic Regression baseline achieves an accuracy of 91.86 percent and an F1-score of 95.69 percent. The BERT model achieves an accuracy of 94.51 percent and an F1-score of 97.03 percent. The results indicate that the BERT model provides better overall performance for the given product review dataset. The BERT model also performs better in identifying negative reviews than the traditional baseline. This demonstrates the potential benefit of using contextual transformer-based representations for product review sentiment classification.

The project also includes functionality for predicting the sentiment of a new product review. When a new review is provided, the review is passed through the BERT tokenizer and then processed by the trained BERT model. The model produces classification scores, which are converted into probabilities using the Softmax function. The class with the highest probability is selected as the predicted sentiment, and the corresponding probability is presented as the confidence score. This allows the trained model to be used not only for evaluating the existing dataset but also for analyzing new customer reviews.

Overall, the background of this project is based on the growing importance of customer reviews, the difficulty of manually analyzing large volumes of textual feedback, and the ability of NLP and machine learning techniques to automate sentiment classification. The project combines a traditional TF-IDF and Logistic Regression approach with a modern BERT-based approach to demonstrate how different NLP methods can be applied to the same sentiment analysis problem. The results show that both approaches can effectively classify product reviews, while BERT provides improved overall performance and demonstrates the advantages of contextual language representations for understanding customer opinions.

1.2 Motivation of the project

The primary motivation for developing this project comes from the increasing importance of online customer reviews in modern e-commerce platforms. Customers frequently depend on the experiences and opinions of previous buyers before making purchasing decisions. Product reviews contain valuable information about customer satisfaction, product quality, performance, reliability, usability, and overall experience. As the number of products and online transactions continues to increase, the amount of customer-generated review data also grows rapidly. Although this data contains useful information, extracting meaningful insights from thousands of individual reviews manually is difficult and time-consuming. This creates a strong need for automated techniques that can analyze large amounts of textual feedback and determine the overall sentiment expressed by customers.

Another important motivation is the difficulty of understanding customer opinions from numerical ratings alone. A rating provides a simple indication of customer satisfaction, but it does not explain why a customer provided a particular rating. For example, two customers may give the same rating while describing completely different experiences in their written reviews. The review text can provide additional information about specific strengths, weaknesses, problems, and expectations that cannot be fully represented by a numerical rating. Therefore, analyzing the written content of reviews along with their associated ratings can provide a deeper understanding of customer sentiment. This project is motivated by the need to use Natural Language Processing techniques to transform this unstructured textual information into meaningful sentiment classifications.

The large volume of reviews also makes manual sentiment identification impractical. If an online product has thousands of reviews, a person would need to read each review individually to determine whether the customer had a positive or negative experience. Such a process requires considerable time and may also introduce inconsistency because different people may interpret the same review differently. An automated machine learning system can process a large number of reviews systematically and provide sentiment predictions much faster. This project therefore aims to demonstrate how machine learning and NLP can be used to automate the process of identifying positive and negative opinions from customer review text.

The project is also motivated by the need to understand how different NLP approaches perform when applied to the same sentiment classification problem. Traditional machine learning methods such as TF-IDF provide an efficient way to represent textual information numerically and can perform well on many text classification tasks. In this project, TF-IDF is combined with Logistic Regression to establish a baseline model. Creating a baseline is important because it provides a reference point for evaluating more advanced approaches. Instead of using only a modern deep learning model, the project compares a traditional NLP pipeline with a transformer-based approach, allowing the differences in their capabilities and performance to be studied.

A further motivation for using BERT is the limitation of traditional word-frequency-based approaches in understanding context. Human language is highly dependent on the relationship between words and the surrounding sentence. The meaning of a statement can change significantly depending on word order, negation, and context. For example, the statements “the product is good” and “the product is not good” contain many of the same words but express opposite opinions. A method based mainly on word frequency may have difficulty representing such contextual differences. BERT provides a contextual representation of language by considering the surrounding words when generating representations. Therefore, using BERT in this project provides an opportunity to investigate whether contextual language understanding can improve product review sentiment classification.

The characteristics of the selected Amazon product review dataset provide another motivation for the study. The dataset contains 4,915 records and includes review text, ratings, review summaries, helpful votes, reviewer information, product identifiers, and review dates. After

examining the data, one record containing missing review text was removed because review text is essential for sentiment classification, resulting in 4,914 usable reviews. The ratings were converted into binary sentiment labels, where ratings of four and five represent positive sentiment and ratings of one, two, and three represent negative sentiment. This transformation creates a practical sentiment classification problem in which the model can learn the relationship between the language used in reviews and the sentiment associated with those reviews.

The class distribution of the dataset provides an additional reason for conducting this study carefully. The final dataset contains 4,448 positive reviews and 466 negative reviews, showing a strong imbalance between the two sentiment classes. A model that simply predicts the majority positive class could achieve relatively high accuracy without effectively identifying negative reviews. Therefore, the project is motivated not only by achieving high overall accuracy but also by examining precision, recall, and F1-score for the individual sentiment classes. This makes the evaluation more meaningful and helps determine whether the models can identify minority-class negative reviews effectively rather than simply relying on the dominant positive class.

The project is also motivated by the educational objective of understanding the complete machine learning workflow for a real-world Natural Language Processing problem. Rather than focusing only on the final prediction, the project covers multiple stages including dataset examination, data cleaning, exploratory analysis, sentiment label creation, train-test splitting, traditional feature extraction, baseline model development, transformer-based model fine-tuning, performance evaluation, and prediction on new reviews. This provides practical experience in applying machine learning concepts to real textual data and demonstrates how different components of an NLP pipeline work together to solve a classification problem.

Another motivation is to evaluate whether the additional complexity of a transformer-based model provides a measurable improvement over a traditional machine learning approach. The TF-IDF and Logistic Regression baseline achieved an accuracy of 91.86 percent and an F1-score of 95.69 percent, while the BERT model achieved an accuracy of 94.51 percent and an F1-score of 97.03 percent. The improvement demonstrates that the contextual representation provided by BERT can be beneficial for the given sentiment classification task. The

comparison therefore provides practical evidence of the difference between a conventional NLP representation and a pretrained transformer-based language model.

The project is further motivated by the possibility of applying the trained model to new customer feedback. After training, the BERT model can receive a new product review, tokenize the input, generate classification scores, and use the Softmax function to obtain probabilities for the sentiment classes. The sentiment with the highest probability can then be presented as the predicted result along with its confidence value. This makes the project more than a theoretical comparison of algorithms because it demonstrates how a trained sentiment classification model can be used to process previously unseen customer feedback.

Finally, the project is motivated by the broader usefulness of automated sentiment analysis for understanding customer feedback at scale. Businesses can potentially use sentiment classification to identify general customer satisfaction trends, detect frequently occurring negative experiences, and understand areas where products or services may require improvement. Customers can also benefit indirectly when organizations use review analysis to improve product quality and customer service. Within the academic scope of this project, the main purpose is to demonstrate the application and comparison of NLP techniques rather than to make business decisions automatically. The project therefore provides a practical case study of how traditional machine learning and modern transformer-based NLP can be applied to a real-world product review sentiment classification problem.

1.3 Overview of the Work Carried Out

The work carried out in this project focuses on developing a Natural Language Processing based system for classifying Amazon product reviews into positive and negative sentiment categories and comparing the performance of a traditional machine learning approach with a modern transformer-based approach. The complete work follows a structured machine learning workflow beginning with understanding the dataset and its attributes, followed by data preparation, exploratory data analysis, sentiment label creation, model development, model evaluation, comparison of results, and prediction of sentiment for new reviews. The overall objective is to investigate how effectively textual customer reviews can be automatically analyzed and classified according to the sentiment expressed by the reviewer.

The first stage of the work involved loading and examining the Amazon product review dataset. The original dataset contained 4,915 records and 12 attributes, including reviewer information, product identifiers, review text, product ratings, review summaries, review dates, and helpful vote information. The dataset was examined to understand its dimensions, data types, available attributes, and overall structure. Missing-value analysis was also performed to identify incomplete records. Two missing values were identified, including one missing reviewer name and one missing review text. Since the review text is the primary input required for sentiment classification, the record with missing review text was removed, resulting in 4,914 usable reviews for the subsequent analysis and model development.

The next stage involved preparing the target sentiment labels from the product ratings. Since the original dataset contains numerical ratings ranging from one to five, these ratings were converted into two sentiment categories to formulate a binary classification problem. Ratings of one, two, and three were classified as negative sentiment, while ratings of four and five were classified as positive sentiment. This transformation allowed the project to use the written review as the input feature and the generated sentiment category as the target variable. The resulting dataset contained 4,448 positive reviews and 466 negative reviews, which also revealed a significant class imbalance that needed to be considered during model evaluation.

Exploratory Data Analysis was then carried out to understand important characteristics of the review dataset before training the models. The analysis examined the distribution of product ratings, the number of reviews belonging to each sentiment category, review length, helpful votes, review dates, frequent words, and relationships between selected numerical variables. The rating distribution showed that five-star reviews represented the largest portion of the dataset, while lower ratings were considerably less frequent. Review length was also examined across different rating categories, showing differences in the average number of words used by customers for different ratings. These exploratory findings helped provide a better understanding of customer review behavior and the characteristics of the textual data used for classification.

After data preparation and exploratory analysis, the dataset was divided into training and testing subsets. An 80 percent training and 20 percent testing strategy was used, with stratification to preserve the distribution of positive and negative sentiment classes in both

subsets. The resulting training dataset contained 3,931 reviews and the testing dataset contained 983 reviews. The training data was used to develop the classification models, while the testing data was kept separate so that the final performance of the trained models could be evaluated using previously unseen reviews.

The first classification approach implemented was a traditional NLP pipeline using TF-IDF and Logistic Regression. TF-IDF was used to convert the textual reviews into numerical feature vectors that could be processed by a machine learning algorithm. The vectorizer was configured with a maximum of 5,000 features, included both unigrams and bigrams, and removed common English stop words. The resulting TF-IDF representation was then provided to a Logistic Regression classifier. This model was used as the baseline because it provides a relatively simple and established approach for text classification and creates a reference point for comparing the performance of the more advanced BERT model.

The second and more advanced approach used the Bidirectional Encoder Representations from Transformers, commonly known as BERT. The pretrained bert-base-uncased model was selected and fine-tuned for binary sentiment classification. A corresponding BERT tokenizer was used to convert review text into tokens that could be processed by the model. The input sequence length was limited to 128 tokens, with longer reviews being truncated and shorter reviews being padded. The pretrained BERT model was then trained on the prepared review dataset using two output classes representing negative and positive sentiment. The model was fine-tuned using a learning rate of 2e-5, a batch size of 8, two training epochs, and a weight decay value of 0.01.

After training both models, their performance was evaluated using multiple classification metrics. Accuracy was used to measure the overall proportion of correctly classified reviews, while precision and recall were used to examine the quality of the model's predictions and its ability to identify the relevant sentiment class. The F1-score was also calculated to provide a combined measure of precision and recall. Because the dataset contained substantially more positive reviews than negative reviews, evaluating multiple metrics was particularly important for understanding whether the models were performing effectively on both classes rather than relying only on overall accuracy.

The evaluation showed that the TF-IDF and Logistic Regression baseline achieved an accuracy of 91.86 percent, with a precision of 91.84 percent, recall of 99.89 percent, and F1-score of 95.69 percent. The BERT model achieved an accuracy of 94.51 percent, precision of 95.14 percent, recall of 98.99 percent, and F1-score of 97.03 percent. The results indicate that BERT provided better overall performance than the traditional TF-IDF and Logistic Regression approach. In particular, the BERT model provided stronger performance for identifying negative reviews, which were the minority class in the dataset. The comparison demonstrates the benefit of using contextual representations from a transformer-based model for the given product review sentiment classification task.

The project also included an analysis of the performance of the models at the individual sentiment-class level. The TF-IDF baseline performed very strongly for the positive class but had substantially lower recall for negative reviews, indicating difficulty in identifying the minority class. The BERT model improved the identification of negative reviews, achieving a negative-class precision of 0.84, recall of 0.52, and F1-score of 0.64, compared with the baseline negative-class recall of 0.15 and F1-score of 0.26. This comparison was important because it demonstrated that the improvement provided by BERT was not limited to overall accuracy but also included better handling of the minority sentiment class.

Finally, the trained BERT model was saved together with its tokenizer so that it could be reused for predicting the sentiment of new product reviews. A prediction function was implemented in which a new review is tokenized and passed through the trained BERT model. The model generates classification logits, which are converted into probability values using the Softmax function. The class with the highest probability is selected as the predicted sentiment, and the corresponding probability is used as the confidence score. This provides a practical way to apply the trained model to previously unseen customer reviews.

Overall, the work carried out in this project represents a complete NLP-based sentiment classification workflow, beginning with raw Amazon review data and progressing through data cleaning, exploratory analysis, sentiment label generation, traditional machine learning, transformer-based deep learning, evaluation, comparison, and new-review prediction. The project demonstrates the difference between a conventional TF-IDF based representation and the contextual language understanding provided by BERT. The final results show that both

approaches can successfully classify product review sentiment, while the BERT-based approach achieved higher accuracy and F1-score and provided better performance in identifying negative reviews.

1,4 Scope and Responsible-Use Statement

The scope of this project is to develop and evaluate a Natural Language Processing based system for automatic sentiment classification of Amazon product reviews. The system focuses on analyzing the textual content of customer reviews and classifying them into two categories, namely positive sentiment and negative sentiment. The project uses product ratings to generate the sentiment labels, where ratings of four and five are treated as positive and ratings of one, two, and three are treated as negative. The main purpose of the system is to demonstrate how machine learning and transformer-based NLP techniques can be applied to a real-world text classification problem and how the performance of different approaches can be compared using standard evaluation metrics.

The project covers the complete workflow required for developing a sentiment analysis model. This includes examining the dataset, identifying and handling missing values, preparing the sentiment labels, performing exploratory data analysis, dividing the data into training and testing sets, developing a traditional TF-IDF and Logistic Regression baseline, and developing a BERT-based sentiment classification model. The project also covers the evaluation of the models using accuracy, precision, recall, and F1-score. The comparison between the baseline and BERT models is an important part of the project because it provides an understanding of the advantages of contextual transformer-based language representations over traditional word-based feature representations.

The scope also includes the use of the trained BERT model for predicting the sentiment of new product reviews. A new review can be provided to the trained model, converted into the required token representation using the BERT tokenizer, and processed to obtain prediction probabilities. The sentiment with the highest probability is selected as the predicted result, while the corresponding probability can be interpreted as the model's confidence for that prediction. This functionality demonstrates how the trained model can be reused for previously

unseen reviews and provides a practical application of the sentiment classification approach developed in the project.

The project can potentially be extended to larger and more diverse review datasets in future work. The current implementation focuses on binary sentiment classification, but the same general approach could be extended to multiple sentiment categories such as positive, neutral, and negative. Future versions could also investigate aspect-based sentiment analysis, where the system identifies sentiment toward specific product characteristics such as price, quality, delivery, design, or performance. Additional improvements could include handling more complex linguistic patterns, increasing the size and diversity of the training data, experimenting with other transformer models, and developing an application interface through which users can submit reviews and receive sentiment predictions.

The scope of the project is also limited by the characteristics of the available dataset. The dataset contains a considerably larger number of positive reviews than negative reviews, resulting in class imbalance. Therefore, the model's performance should not be interpreted only from overall accuracy. Precision, recall, and F1-score are also considered to obtain a more complete understanding of classification performance, particularly for the minority negative class. The project also uses a maximum BERT sequence length of 128 tokens, meaning that reviews longer than this limit are truncated during model processing. Consequently, information occurring beyond the selected sequence length may not be considered by the model.

The project should be used responsibly because sentiment predictions generated by machine learning models are not guaranteed to be completely correct. The model learns patterns from the available training data and may produce incorrect predictions for reviews containing sarcasm, ambiguous language, unusual expressions, spelling errors, or contextual meanings that are difficult to interpret. The confidence value produced by the model should therefore not be treated as absolute certainty. A high confidence score represents the model's estimated probability for a class based on its learned patterns and does not guarantee that the predicted sentiment accurately represents the reviewer's actual intention.

Responsible use also requires recognizing the limitations of the dataset and avoiding unsupported conclusions about customers or products. The model is designed to classify the sentiment of review text and should not be used to make sensitive decisions about individual people. The output should be interpreted as an automated analysis of textual content rather than as a definitive judgment about a customer, reviewer, business, or product. When the system is used in a practical environment, human review may still be necessary, especially when predictions could influence important business decisions or when the review contains unclear or highly contextual language.

Data privacy and responsible handling of review information are also important considerations. Although the project uses an existing product review dataset for academic analysis, any real-world deployment should ensure that customer information is handled appropriately and that unnecessary personal information is not collected or exposed. Information such as reviewer identifiers should not be used unnecessarily for sentiment prediction. The focus of the system should remain on the review content required for the intended analytical task, while appropriate privacy and data-protection practices should be followed when working with real customer information.

The project is intended primarily for academic learning, experimentation, and demonstration of NLP and machine learning concepts. The comparison between TF-IDF with Logistic Regression and BERT is intended to demonstrate the differences between traditional and modern NLP approaches rather than to claim that one model will always perform better on every dataset. The reported performance is specific to the dataset, preprocessing steps, model configurations, training procedure, and evaluation method used in this study. Therefore, the results should not automatically be generalized to all product categories, customer populations, or review datasets.

Overall, the scope of this project is to demonstrate an end-to-end sentiment analysis system using Amazon product reviews, traditional NLP techniques, and a transformer-based BERT model. The project provides a foundation for automated customer feedback analysis while recognizing the importance of model limitations, dataset characteristics, privacy, fairness, and human oversight. Responsible use of the developed system requires treating its predictions as

analytical assistance rather than unquestionable decisions and ensuring that the system is evaluated carefully before being used in any real-world application.

PROBLEM STATEMENT

2.1 Case-Study Scenario

The case study focuses on the problem of automatically analyzing customer opinions expressed through online product reviews. In modern e-commerce platforms, customers frequently provide written feedback after purchasing products. These reviews can contain valuable information about product quality, performance, usability, reliability, price, and overall satisfaction. At the same time, customers can provide numerical ratings that indicate their overall experience. As the number of products and online transactions increases, the number of customer reviews also increases, creating a large volume of textual information that can be difficult to analyze manually. The case study therefore considers how Natural Language Processing and machine learning techniques can be used to automatically identify the sentiment expressed in these reviews.

For this case study, an Amazon product review dataset is considered. The original dataset contains 4,915 review records with 12 attributes, including reviewer information, product identifiers, review text, ratings, review summaries, review dates, and helpful vote information. The review text represents the main source of information for sentiment analysis because it contains the customer's written opinion about the product. The numerical rating associated with each review is used to create the target sentiment category. One record containing missing review text is removed because a review without textual content cannot be used effectively for text-based sentiment classification. After this preparation step, 4,914 usable reviews are available for analysis.

The case study converts the original five-level rating information into a binary sentiment classification problem. Reviews with ratings of four and five are considered positive, while reviews with ratings of one, two, and three are considered negative. This transformation provides a clear classification objective in which the system receives the review text as input and predicts whether the expressed sentiment is positive or negative. After the transformation,

the dataset contains 4,448 positive reviews and 466 negative reviews. The significant difference between these two classes represents an important characteristic of the case study because the model must be evaluated carefully to determine whether it can identify negative reviews rather than simply predicting the majority positive class.

Before developing the sentiment classification models, exploratory data analysis is performed to understand the characteristics of the dataset. The analysis examines the distribution of ratings, sentiment categories, review lengths, helpful votes, review dates, frequently occurring words, and relationships between selected numerical variables. The analysis shows that five-star reviews form the largest portion of the dataset, while lower ratings occur much less frequently. The review-length analysis also indicates that reviews associated with lower ratings tend to have different average lengths compared with higher-rated reviews. These observations provide useful context for understanding the nature of the customer feedback used in the case study.

The dataset is divided into training and testing subsets using an 80 percent training and 20 percent testing split. Stratification is applied to maintain the relative distribution of positive and negative reviews in both subsets. The resulting training set contains 3,931 reviews and the testing set contains 983 reviews. The training set is used to learn patterns associated with positive and negative sentiment, while the testing set is reserved for evaluating the models on reviews that were not used during training. This separation allows the case study to assess how well the developed models generalize to previously unseen review text.

Two different approaches are considered in the case study to solve the sentiment classification problem. The first approach is a traditional Natural Language Processing pipeline based on TF-IDF and Logistic Regression. TF-IDF is used to convert the review text into numerical feature representations by considering the importance of words and word combinations within the review collection. The resulting features are provided to a Logistic Regression classifier, which learns to distinguish between positive and negative reviews. This approach serves as the baseline model and provides a reference for evaluating the performance of the more advanced approach.

The second approach uses BERT, which is a transformer-based language model capable of generating contextual representations of text. The pretrained bert-base-uncased model is fine-tuned for binary sentiment classification using the prepared Amazon review dataset. The review text is tokenized using the BERT tokenizer, with a maximum sequence length of 128 tokens. Longer reviews are truncated and shorter reviews are padded before being provided to the model. The model is trained to produce two output classes corresponding to negative and positive sentiment. Through this approach, the case study examines whether contextual language representations can improve sentiment classification compared with the traditional TF-IDF-based approach.

The performance of both approaches is evaluated using accuracy, precision, recall, and F1-score. These measures are particularly important in this case study because the dataset contains a strong imbalance between positive and negative reviews. The TF-IDF and Logistic Regression model achieves an accuracy of 91.86 percent and an F1-score of 95.69 percent. The BERT model achieves an accuracy of 94.51 percent and an F1-score of 97.03 percent. The results indicate that the BERT-based approach provides better overall performance for the selected dataset. The BERT model also provides substantially better identification of negative reviews compared with the traditional baseline, demonstrating the value of evaluating individual sentiment classes rather than relying only on overall accuracy.

The case study also includes the practical application of the trained BERT model to new product reviews. A new review can be provided as input to the prediction function, where the text is first tokenized and then passed through the trained BERT model. The model generates classification scores, which are converted into probabilities using the Softmax function. The sentiment with the highest probability is selected as the prediction, and the associated probability is returned as the confidence value. This demonstrates how the developed model can be applied to previously unseen customer feedback after the training and evaluation stages have been completed.

Therefore, the central scenario of this case study is the automated analysis of Amazon customer reviews using NLP and machine learning. The study compares a traditional TF-IDF and Logistic Regression approach with a BERT-based transformer approach to determine how effectively each method can classify customer reviews into positive and negative sentiment

categories. The scenario provides a practical application of machine learning concepts to a real-world textual dataset and demonstrates the complete process from data preparation and exploratory analysis to model development, evaluation, comparison, and prediction of new reviews.

2.2 Central Case-Study Question

The central question of this case study is: How effectively can Natural Language Processing and machine learning techniques automatically classify Amazon product reviews into positive and negative sentiment categories, and does a transformer-based BERT model provide better sentiment classification performance than a traditional TF-IDF and Logistic Regression approach?

This question forms the foundation of the entire study because the project is designed not only to classify customer reviews but also to compare two different approaches to sentiment analysis. The first approach uses TF-IDF to represent review text as numerical features and Logistic Regression to perform sentiment classification. The second approach uses the pretrained BERT transformer model, which is capable of learning contextual relationships between words in a review. By applying both approaches to the same prepared dataset and evaluating them using the same testing data, the study can determine how their classification performance differs.

The question also considers whether the models are able to identify both positive and negative customer opinions effectively. This is particularly important because the dataset contains substantially more positive reviews than negative reviews. Therefore, overall accuracy alone may not provide a complete representation of model performance. Precision, recall, and F1-score are also considered to determine how effectively the models classify the different sentiment categories, especially the minority negative class.

The case study further considers whether the trained BERT model can be used to analyze previously unseen product reviews. After training and evaluation, the model is used to predict the sentiment of a new review and provide a corresponding confidence value. This demonstrates the practical application of the developed sentiment analysis system beyond the original dataset.

Thus, the central case-study question investigates both the effectiveness of automated sentiment classification and the comparative advantage of contextual transformer-based NLP over a traditional text-feature-based machine learning approach. The answer to this question is determined through systematic data preparation, exploratory analysis, model implementation, performance evaluation, and comparison of the TF-IDF with Logistic Regression baseline and the BERT-based sentiment classification model.

2.3 Formal Problem Definition

The formal problem addressed in this project is the automatic classification of customer product reviews into positive and negative sentiment categories using Natural Language Processing and machine learning techniques. Given a collection of Amazon product reviews containing textual review information and associated product ratings, the objective is to develop a computational model that can learn the relationship between the language used in a review and the sentiment represented by that review. The problem is formulated as a supervised binary text classification task, where the review text is used as the input and the sentiment category derived from the product rating is used as the target output.

Let the prepared dataset be represented as (D = {(x_i,y_i)}_{i=1}^{N}), where (x_i) represents the textual content of the (i^{th}) product review and (y_i) represents its corresponding sentiment label. The sentiment label is defined as (y_i = 0) for negative sentiment and (y_i = 1) for positive sentiment. The labels are generated from the original product ratings, with ratings of one, two, and three assigned to the negative class and ratings of four and five assigned to the positive class. After removing the record with missing review text, the prepared dataset contains 4,914 usable reviews, consisting of 4,448 positive reviews and 466 negative reviews.

The objective of the classification problem is to learn a function (f(x)) that maps a review text (x) to its predicted sentiment class (\hat{y}). The learned function should identify linguistic patterns associated with positive and negative customer opinions and should generalize these patterns to reviews that were not included during model training. In mathematical form, the desired prediction can be represented as (\hat{y}=f(x)), where (\hat{y}) belongs to the set of possible sentiment classes ({0,1}). The quality of this function is determined by comparing its predictions with the actual sentiment labels in the testing dataset.

The problem is addressed using two different modeling approaches. The first approach represents the review text using Term Frequency–Inverse Document Frequency, or TF-IDF, and then applies Logistic Regression for classification. The TF-IDF representation converts the textual information into numerical feature vectors based on the importance of words and word combinations within the review collection. Logistic Regression then learns a decision function that separates the positive and negative review representations. This model serves as the traditional machine learning baseline for the study.

The second approach uses a pretrained Bidirectional Encoder Representations from Transformers, or BERT, model. In this approach, the review text is tokenized using the BERT tokenizer and converted into input representations suitable for the transformer model. The bert-base-uncased model is fine-tuned for two-class sequence classification so that it can learn contextual relationships within product reviews and predict whether a review belongs to the positive or negative sentiment class. A maximum sequence length of 128 tokens is used, with longer reviews being truncated and shorter reviews being padded.

The prepared dataset is divided into training and testing subsets using an 80 percent to 20 percent split with stratification. The training subset contains 3,931 reviews and is used to learn the parameters of the classification models, while the testing subset contains 983 reviews and is used to evaluate their performance on unseen data. The use of a separate testing dataset is necessary to determine whether the learned models can generalize beyond the reviews used during training.

An important characteristic of the formal problem is the imbalance between the two sentiment classes. The prepared dataset contains 4,448 positive reviews compared with only 466 negative reviews. This imbalance creates a potential challenge because a model may obtain high overall accuracy by predicting the majority positive class while performing poorly on negative reviews. Therefore, the problem is not simply to maximize the number of correct predictions but to develop a model that provides meaningful classification performance across both sentiment categories. For this reason, accuracy, precision, recall, and F1-score are used to evaluate the developed models.

The formal problem also includes the comparison of the two modeling approaches to determine whether contextual representations provided by BERT result in improved classification performance compared with the traditional TF-IDF and Logistic Regression approach. The TF-IDF and Logistic Regression model achieves an accuracy of 91.86 percent and an F1-score of 95.69 percent, whereas the BERT model achieves an accuracy of 94.51 percent and an F1-score of 97.03 percent. The comparison therefore provides an empirical basis for determining the effectiveness of the two approaches on the selected product review dataset.

Finally, the problem includes the ability to apply the trained BERT model to a previously unseen review. Given a new review text (x_{new}), the trained model produces classification scores that are converted into probabilities using the Softmax function. The class with the highest probability is selected as the predicted sentiment. Therefore, the overall formal problem can be stated as the development of an NLP-based binary classification system that learns from labeled Amazon product reviews, accurately distinguishes positive and negative sentiment, evaluates the performance of traditional and transformer-based approaches, and applies the trained model to new customer review text.

2.4 Objectives that Guide the Study

The primary objective of this study is to develop an automated sentiment analysis system capable of classifying Amazon product reviews into positive and negative sentiment categories using Natural Language Processing and machine learning techniques. The study aims to use the textual content of customer reviews as the primary input and derive sentiment labels from the associated product ratings. By converting the original rating information into binary

sentiment categories, the study establishes a supervised classification problem that can be systematically addressed using machine learning models.

A major objective of the study is to understand and prepare the Amazon product review dataset before applying machine learning techniques. This includes examining the structure of the dataset, identifying missing values, handling the record containing missing review text, understanding the available attributes, and preparing the data required for sentiment classification. The study aims to ensure that the review data is in an appropriate form for subsequent exploratory analysis and model development.

Another objective is to perform exploratory data analysis to understand the characteristics and patterns present in the product review dataset. The study examines the distribution of product ratings and sentiment classes, review length, helpful votes, review dates, frequently occurring words, and relationships between selected numerical variables. This analysis is intended to provide an understanding of the dataset before model training and to identify characteristics such as the significant imbalance between positive and negative reviews.

The study also aims to develop a traditional Natural Language Processing baseline using TF-IDF and Logistic Regression. TF-IDF is used to transform the textual reviews into numerical representations based on word and word-combination importance, while Logistic Regression is used to classify the resulting representations into positive and negative sentiment categories. Establishing this baseline provides a reference against which the performance of a more advanced transformer-based model can be measured.

A further objective is to implement a BERT-based sentiment classification model using the pretrained bert-base-uncased transformer. The study aims to fine-tune BERT on the prepared product review dataset so that the model can learn contextual relationships within customer reviews and distinguish between positive and negative sentiment. The study uses BERT tokenization with a maximum sequence length of 128 tokens and applies the trained model to the binary sentiment classification task.

Another important objective is to evaluate and compare the performance of the traditional and transformer-based approaches using appropriate classification metrics. The study evaluates the models using accuracy, precision, recall, and F1-score rather than relying only on accuracy. This is particularly important because the dataset contains 4,448 positive reviews and only 466 negative reviews. Evaluating multiple metrics allows the study to determine whether the models are able to identify both sentiment classes effectively, especially the minority negative class.

The study also aims to determine whether the contextual language representations provided by BERT offer an improvement over the traditional TF-IDF and Logistic Regression approach. The performance of both models is evaluated using the same testing dataset so that their results can be compared consistently. The TF-IDF and Logistic Regression model achieves an accuracy of 91.86 percent and an F1-score of 95.69 percent, while the BERT model achieves an accuracy of 94.51 percent and an F1-score of 97.03 percent. These results are used to assess the effectiveness of the two approaches for the selected sentiment classification problem.

Another objective is to examine the ability of the trained BERT model to classify previously unseen product reviews. The study implements a prediction process in which a new review is tokenized, processed by the trained BERT model, and converted into sentiment probabilities using the Softmax function. The sentiment with the highest probability is selected as the predicted class, and the corresponding probability is used as the confidence value. This objective demonstrates the practical use of the developed model beyond evaluation on the existing test dataset.

Finally, the study aims to demonstrate an end-to-end application of Natural Language Processing and machine learning to a real-world customer feedback problem while recognizing the limitations of the dataset and the developed models. The study seeks to show how customer review text can be transformed into useful sentiment information, how traditional and transformer-based NLP methods can be compared, and how a trained sentiment classification model can be applied to new review data. Through these objectives, the study provides a structured evaluation of automated product review sentiment analysis using both conventional machine learning and modern BERT-based NLP techniques.

2.5 Assumptions and Limitations

The study is based on several assumptions regarding the Amazon product review dataset and the sentiment classification task. The first assumption is that the written review text contains sufficient information to determine the general sentiment of the customer toward the product. The study assumes that the language used in a review reflects the reviewer's overall opinion and that this opinion can be classified into either a positive or negative category. The product rating associated with each review is assumed to provide a reasonable basis for generating the sentiment labels used as the target variable.

The study assumes that ratings of four and five represent positive sentiment, while ratings of one, two, and three represent negative sentiment. This assumption is necessary to convert the original five-level rating system into a binary classification problem. Under this approach, the rating is treated as an indicator of the overall sentiment expressed by the customer. However, this conversion does not necessarily mean that every review with a particular rating contains exactly the same emotional meaning, since customers may express different opinions even when assigning the same numerical rating.

Another assumption is that the available Amazon review dataset is sufficiently representative for demonstrating the sentiment classification problem addressed in this academic study. The dataset contains 4,915 original records and, after removing one record with missing review text, 4,914 usable reviews. The study assumes that these reviews provide enough textual variation for the models to learn useful patterns associated with positive and negative sentiment. However, the results are considered specific to the selected dataset and should not automatically be assumed to represent all Amazon reviews or all types of customer feedback.

The study also assumes that the preprocessing and data preparation steps are appropriate for the selected classification task. The record containing missing review text is removed because the review text is required as the main input to the NLP models. The remaining review text is assumed to contain sufficient usable information for feature extraction and model training. For the BERT model, the review text is tokenized and processed using a maximum sequence length of 128 tokens, with longer sequences truncated and shorter sequences padded.

A further assumption is that the training and testing procedure provides a reasonable estimate of model performance on unseen data. The dataset is divided using an 80 percent training and 20 percent testing split with stratification, resulting in 3,931 training reviews and 983 testing reviews. The study assumes that keeping the testing data separate from model training provides an appropriate basis for evaluating generalization. The testing results are therefore treated as an estimate of how the trained models may perform on previously unseen reviews from a similar data distribution.

One of the major limitations of the study is the significant imbalance between the two sentiment classes. The prepared dataset contains 4,448 positive reviews and only 466 negative reviews. This means that positive reviews greatly outnumber negative reviews and can influence the overall classification results. A model may obtain high accuracy by performing very well on the majority positive class while still having difficulty identifying negative reviews. This limitation is particularly visible in the TF-IDF and Logistic Regression baseline, which achieves very high recall for positive reviews but considerably lower recall for negative reviews. Therefore, accuracy alone cannot fully represent the effectiveness of the models.

Another limitation is that the sentiment labels are derived from numerical ratings rather than being independently annotated from the actual review text. Although ratings of one to three are treated as negative and ratings of four and five as positive, a numerical rating may not always perfectly represent the exact sentiment expressed in the written review. A customer may provide a high rating while mentioning a minor negative issue, or provide a lower rating while still describing some positive aspects. Consequently, the generated labels may contain some degree of noise or may not capture all the nuances of the reviewer's opinion.

The study is also limited by the ability of the models to understand complex language. Product reviews may contain sarcasm, irony, ambiguous expressions, spelling mistakes, informal language, or statements whose meaning depends heavily on context. Although BERT provides contextual language representations and performs better than the traditional baseline in this study, it cannot be assumed that every prediction is correct. The model may still misclassify reviews when the sentiment is subtle or difficult to determine from the available text.

Another limitation is the maximum sequence length used for the BERT model. The implementation uses a maximum length of 128 tokens. Reviews longer than this limit are truncated before being provided to the model. As a result, information appearing after the maximum sequence length may not be considered during prediction. This limitation can affect sentiment classification when important opinions or conclusions occur toward the end of a long review.

The scope of the classification task is another limitation. The current implementation performs binary sentiment classification and distinguishes only between positive and negative sentiment. It does not explicitly classify neutral sentiment or multiple levels of emotional intensity. Similarly, the system does not perform aspect-based sentiment analysis, meaning that it does not separately determine sentiment toward specific characteristics such as price, quality, delivery, design, or performance. Therefore, the model provides an overall sentiment classification rather than a detailed analysis of individual product aspects.

The study is also limited by the size and characteristics of the available dataset. Although 4,914 usable reviews provide sufficient data for demonstrating the approach, the dataset is relatively limited compared with the very large collections of reviews available on modern e-commerce platforms. The data may also represent particular products, customers, writing styles, and review patterns. Consequently, the performance obtained in this study may change when the model is applied to reviews from different products, platforms, domains, or customer populations.

Another limitation concerns the comparison between the traditional and transformer-based models. The TF-IDF and Logistic Regression approach and the BERT approach use fundamentally different text representations and learning mechanisms. The TF-IDF model is comparatively lightweight, while BERT is a pretrained transformer model that requires greater computational resources and training time. Therefore, the comparison primarily focuses on classification performance rather than computational efficiency, deployment cost, or resource requirements. The higher performance of BERT should therefore be considered together with its additional computational complexity.

The reported model performance is also dependent on the specific configuration used in the study. The TF-IDF model uses a maximum of 5,000 features, considers unigrams and bigrams, and uses English stop-word removal. The BERT model uses the pretrained bert-base-uncased architecture, a maximum sequence length of 128 tokens, a learning rate of 2e-5, a batch size of 8, two training epochs, and a weight decay of 0.01. Different preprocessing methods, model architectures, hyperparameters, training durations, or datasets could produce different results. Therefore, the reported accuracy, precision, recall, and F1-score should be interpreted within the specific experimental setup of this study.

Finally, the model predictions should not be treated as completely reliable or as a replacement for human judgment in situations where sentiment classification could influence important decisions. The confidence value generated by the BERT model represents the model's estimated probability for its prediction and should not be interpreted as a guarantee of correctness. The developed system is primarily intended for academic demonstration and analysis of customer review sentiment. Real-world deployment would require further validation using larger and more diverse datasets, additional testing for robustness, careful handling of customer information, and appropriate human oversight.

Overall, the assumptions provide the foundation required to formulate and evaluate the sentiment classification problem, while the limitations define the boundaries within which the results should be interpreted. The study assumes that review ratings provide useful sentiment labels and that the selected dataset is suitable for demonstrating the classification task. At the same time, factors such as class imbalance, rating-based labeling, contextual language, sequence-length restrictions, binary classification, dataset size, and computational differences between models limit the extent to which the results can be generalized. Recognizing these assumptions and limitations is essential for interpreting the findings of the study responsibly and for identifying possible directions for future improvement.

DATASET AND EXPLORATORY DATA ANALYSIS

3.1 About the Dataset

The dataset used in this project is an Amazon product review dataset containing customer feedback about products purchased through an online platform. The dataset provides both textual and numerical information associated with customer reviews, making it suitable for Natural Language Processing and sentiment analysis. The main purpose of using this dataset is to analyze the written opinions of customers and determine whether the expressed sentiment is positive or negative. The dataset contains information related to individual reviewers, products, review text, ratings, review summaries, review dates, and helpfulness information. These attributes provide useful context for understanding the characteristics of customer feedback and for performing exploratory analysis before developing the sentiment classification models.

The original dataset contains 4,915 records and 12 attributes. The attributes include reviewerID, asin, reviewerName, helpful, reviewText, overall, summary, unixReviewTime, reviewTime, day_diff, helpful_yes, and total_vote. The reviewerID identifies the reviewer associated with a particular review, while asin represents the product identifier. The reviewerName contains the name associated with the reviewer when available. The helpful attribute contains information related to the helpfulness of the review. The reviewText attribute contains the actual written review provided by the customer and is the primary textual feature used for sentiment analysis. The overall attribute represents the numerical product rating given by the customer and is used to generate the sentiment labels for the classification task.

The summary attribute contains a short summary provided by the reviewer and can provide additional information about the reviewer's opinion. The unixReviewTime and reviewTime attributes provide information about when the review was submitted, while day_diff represents the relevant difference in days available in the dataset. The helpful_yes attribute represents the number of helpful votes received by a review, and total_vote represents the total number of votes associated with the review. These attributes are useful for exploratory analysis because they provide information about review activity and how other users interacted with the posted reviews.

The dataset was first examined to identify missing values and understand the quality of the available information. A total of two missing values were identified in the original dataset, with one missing value in the reviewerName attribute and one missing value in the reviewText attribute. Since the written review is essential for the NLP-based sentiment classification task, the record with missing review text was removed. The missing reviewer name did not directly affect the sentiment classification because reviewer identity is not required for predicting sentiment. After removing the record without review text, the dataset contained 4,914 usable reviews for subsequent analysis and model development.

The overall rating attribute is particularly important in this project because it is used to transform the original rating information into binary sentiment labels. Ratings of one, two, and three are assigned to the negative sentiment class, represented by the label zero, while ratings of four and five are assigned to the positive sentiment class, represented by the label one. This transformation converts the original rating-based dataset into a supervised binary sentiment classification dataset. The resulting dataset contains 4,448 positive reviews and 466 negative reviews. Therefore, the dataset has a substantial imbalance between positive and negative sentiment categories, with positive reviews representing the majority of the available observations.

The rating distribution further demonstrates the characteristics of the dataset. Five-star reviews form the largest group, containing 3,922 reviews, which represents approximately 79.80 percent of the usable dataset. Four-star reviews contain 527 reviews, representing approximately 10.72 percent. Three-star reviews contain 142 reviews, representing approximately 2.89 percent, while two-star reviews contain 80 reviews, representing approximately 1.63 percent. One-star reviews contain 244 reviews, representing approximately 4.96 percent. This distribution indicates that customers in the dataset were considerably more likely to provide high ratings than low ratings, which explains the strong imbalance observed after converting the ratings into positive and negative sentiment categories.

The review text is the most important attribute for the main objective of this project because it contains the natural language written by customers. The textual reviews vary in length and writing style, with customers using different words, expressions, and sentence structures to communicate their experiences. The exploratory analysis also examines review length across

different rating groups. The average review word count is approximately 103.86 words for one-star reviews, 114.48 words for two-star reviews, 73.25 words for three-star reviews, 56.21 words for four-star reviews, and 44.21 words for five-star reviews. These differences indicate that lower-rated reviews in this dataset tend to contain more words on average, while higher-rated reviews tend to be shorter.

The helpfulness-related attributes provide additional information about customer interaction with the reviews. The average number of helpful votes and total votes varies across the different rating categories. One-star reviews have an average of approximately 8.97 helpful votes and 10.40 total votes, while two-star reviews have approximately 0.24 helpful votes and 0.54 total votes. Three-star reviews have approximately 0.35 helpful votes and 0.54 total votes, four-star reviews have approximately 0.21 helpful votes and 0.61 total votes, and five-star reviews have approximately 1.04 helpful votes and 1.15 total votes. These values are useful during exploratory analysis because they provide information about how review ratings and review helpfulness are distributed within the dataset.

For model development, the cleaned dataset is divided into training and testing subsets using an 80 percent to 20 percent split. Stratified splitting is applied so that the relative distribution of positive and negative sentiment classes is maintained in both subsets. The resulting training dataset contains 3,931 reviews, while the testing dataset contains 983 reviews. The training data is used to learn sentiment patterns from the review text, whereas the testing data is kept separate and used to evaluate the models on previously unseen reviews.

The dataset is used to develop and compare two sentiment classification approaches. The first approach uses TF-IDF to convert review text into numerical features and Logistic Regression to classify the reviews. The second approach uses the pretrained bert-base-uncased transformer model, which is fine-tuned for binary sentiment classification. Therefore, the dataset serves as the common basis for comparing a traditional NLP and machine learning approach with a modern transformer-based approach.

Overall, the Amazon product review dataset is appropriate for this case study because it combines written customer opinions with numerical ratings and additional review-related information. The dataset provides sufficient examples for demonstrating data preparation,

exploratory analysis, sentiment labeling, traditional NLP-based classification, BERT-based classification, and model comparison. At the same time, its strong positive-class imbalance, rating-based sentiment labeling, and variation in review language are important characteristics that must be considered when interpreting the results of the sentiment analysis models.

3.2 Purpose of EDA

The main purpose of Exploratory Data Analysis, or EDA, in this project is to understand the structure, quality, distribution, and important characteristics of the Amazon product review dataset before applying Natural Language Processing and machine learning techniques. EDA provides an initial understanding of the available data and helps determine whether the dataset is suitable for the intended sentiment classification task. Since the project uses both textual and numerical information, exploratory analysis is important for examining the available attributes, identifying potential data-quality issues, understanding customer rating patterns, and determining characteristics that may influence the subsequent modeling process.

An important purpose of EDA is to examine the overall structure of the dataset. The analysis begins by understanding the number of records and attributes, the types of variables available, and the information contained in each attribute. The original dataset contains 4,915 records and 12 attributes, including reviewer information, product identifiers, review text, ratings, review summaries, review dates, and helpfulness-related information. Examining these attributes helps identify which variables are relevant to the sentiment analysis task and which variables are mainly useful for supporting exploratory analysis.

Another purpose of EDA is to identify missing or incomplete information before model development. Missing values can affect data quality and may prevent certain records from being used effectively. In this dataset, two missing values were identified, including one missing reviewerName value and one missing reviewText value. Since the review text is the primary input required for NLP-based sentiment classification, the record with missing review text was removed. This resulted in 4,914 usable reviews. Identifying this issue during EDA ensures that the subsequent modeling process is performed using appropriate input data.

EDA is also performed to understand the distribution of product ratings. The overall rating attribute contains values from one to five and provides an important indication of customer satisfaction. Examining the rating distribution shows that five-star reviews form the largest portion of the dataset, while lower ratings occur considerably less frequently. This information is important because the ratings are later converted into binary sentiment labels, with ratings of one, two, and three treated as negative and ratings of four and five treated as positive. Therefore, understanding the original rating distribution helps explain the resulting sentiment-class distribution.

A further purpose of EDA is to identify class imbalance in the sentiment labels. After converting the ratings into positive and negative categories, the dataset contains 4,448 positive reviews and 466 negative reviews. This imbalance is an important characteristic because it can influence the behavior and evaluation of classification models. A model could achieve high overall accuracy by predicting the majority positive class while performing poorly on negative reviews. Recognizing this issue during EDA motivates the use of multiple evaluation metrics, including precision, recall, and F1-score, rather than relying only on accuracy.

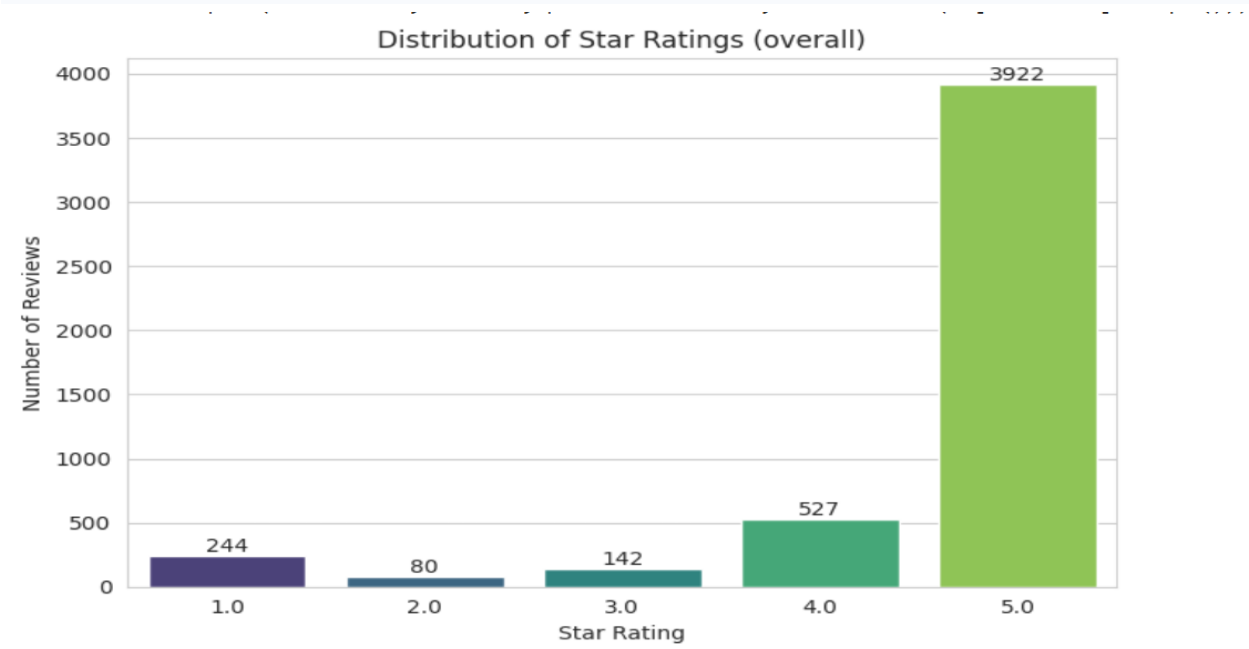
The analysis also aims to understand the characteristics of the review text. Review length is examined to determine how the amount of textual information varies across different rating categories. The average review word count is approximately 103.86 words for one-star reviews, 114.48 words for two-star reviews, 73.25 words for three-star reviews, 56.21 words for four-star reviews, and 44.21 words for five-star reviews. These observations provide insight into differences in customer writing behavior and help demonstrate the variation present in the textual data that will be processed by the NLP models.

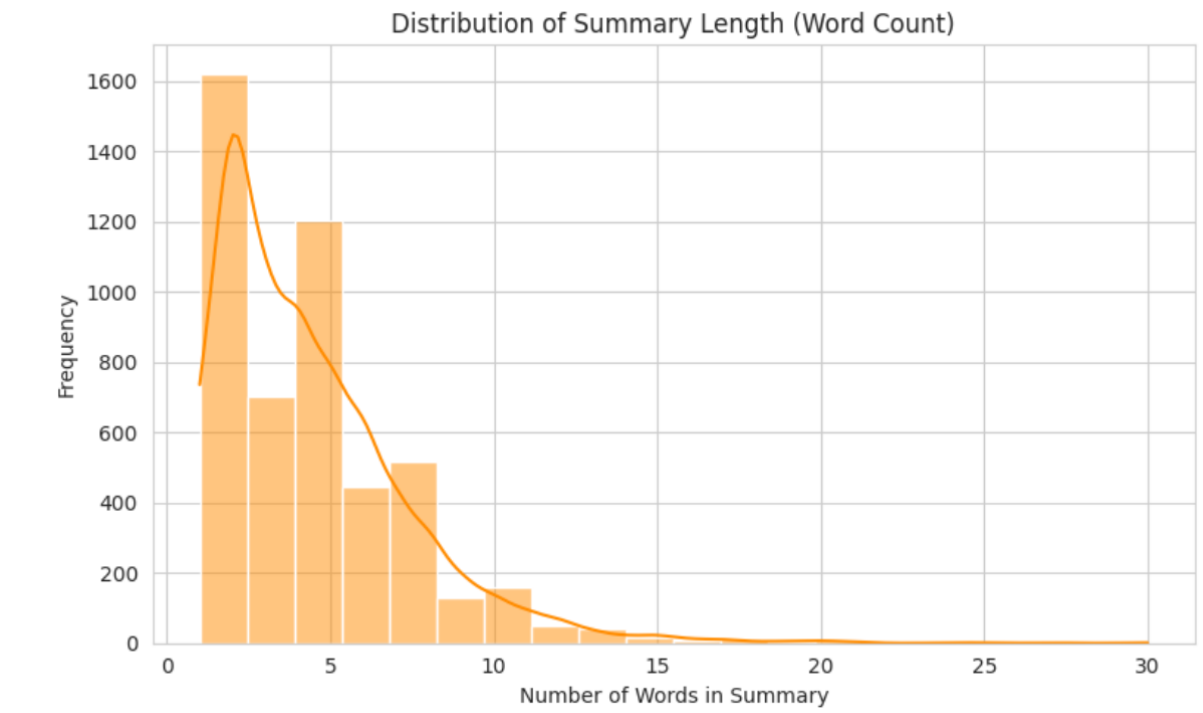
Another purpose of EDA is to examine the helpfulness-related information associated with reviews. The helpful_yes and total_vote attributes provide information about how other users interacted with the reviews. Examining these variables helps provide additional context about the review dataset and allows the distribution of helpful votes to be compared across rating categories. Although these attributes are not the primary input features for the sentiment classification models, they contribute to a broader understanding of the customer review data. EDA is also intended to identify patterns and relationships that may be useful for interpreting the dataset. The analysis includes examining frequently occurring words and relationships

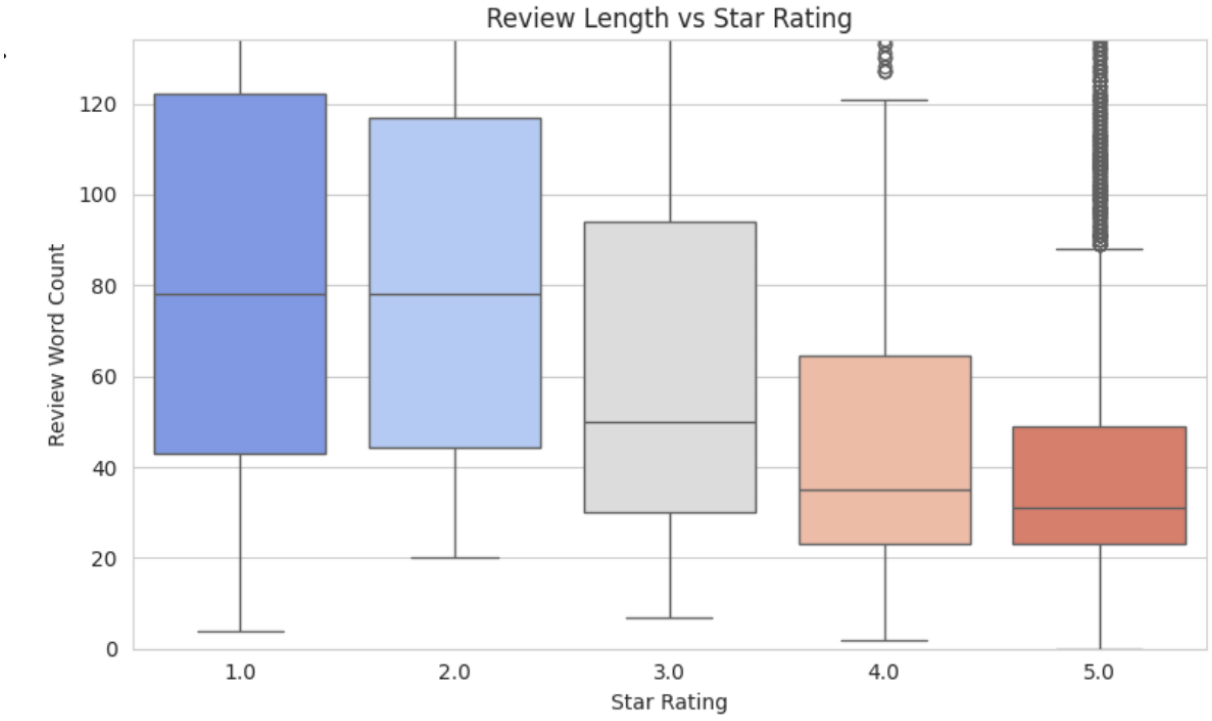
among selected numerical variables. These explorations help provide a better understanding of the language and characteristics present in customer feedback. In an NLP project, such analysis is particularly useful because the textual content is the main source from which sentiment patterns must be learned.

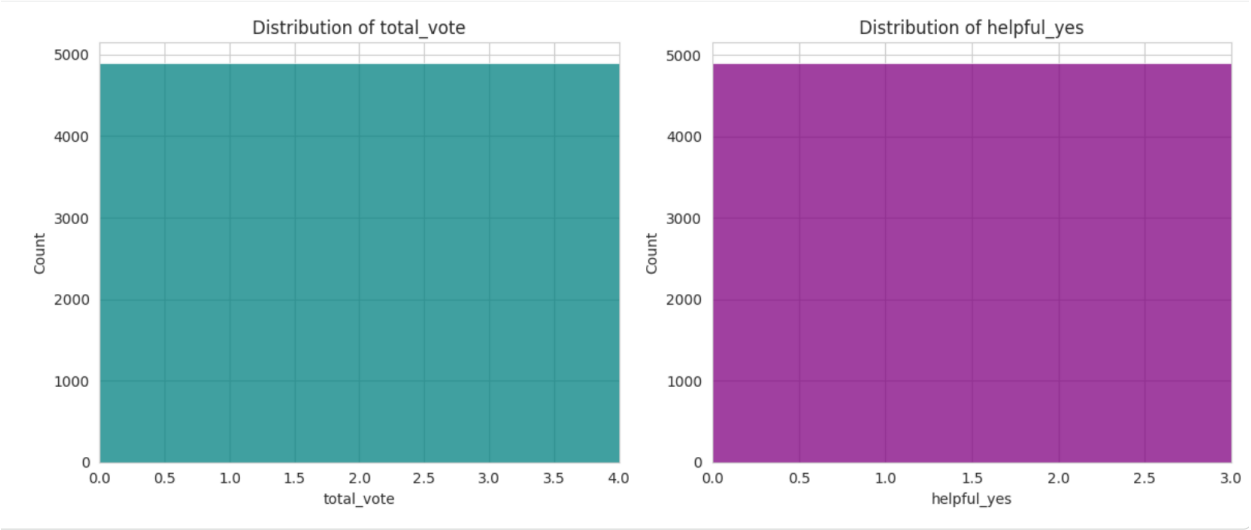
Another important purpose of EDA is to support appropriate decisions during data preparation and model development. By understanding the dataset before training, it becomes possible to determine how missing information should be handled, how the sentiment labels should be created, how the data should be divided for training and testing, and which characteristics require particular attention during evaluation. For example, identifying the strong imbalance between positive and negative reviews supports the decision to use stratified train-test splitting so that both subsets maintain a similar class distribution.

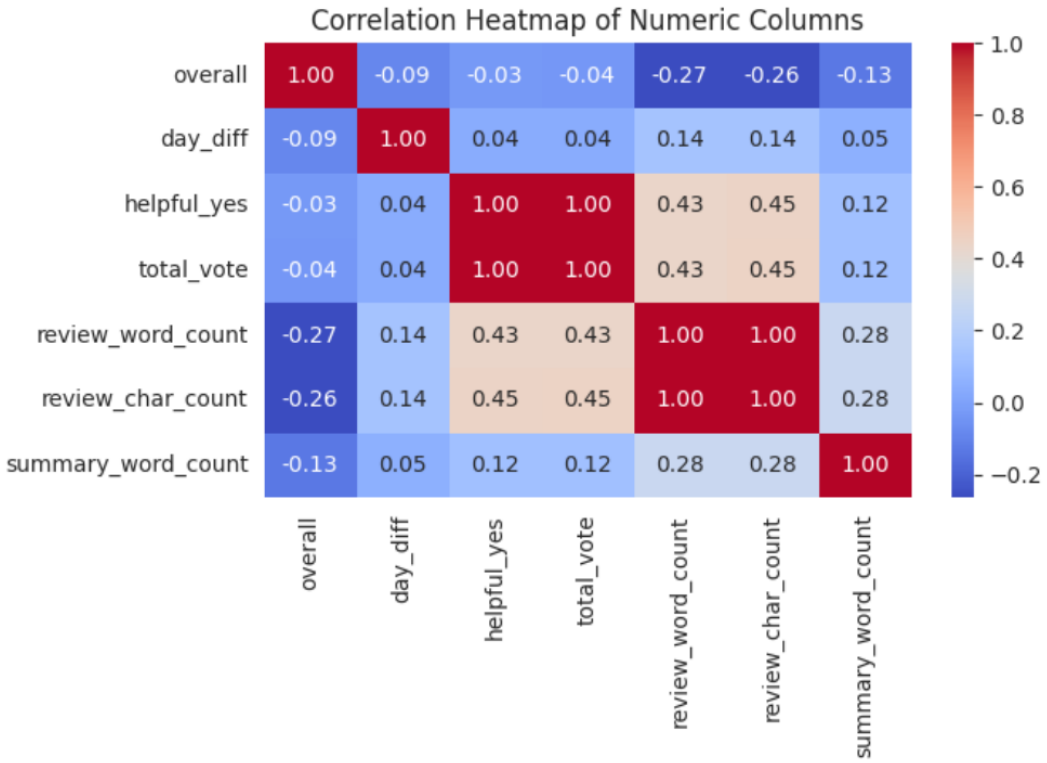
Overall, the purpose of EDA in this project is to establish a clear understanding of the Amazon product review dataset before applying TF-IDF with Logistic Regression and the BERT-based sentiment classification model. EDA helps verify data quality, understand rating and sentiment distributions, identify class imbalance, examine review-length characteristics, analyze helpfulness information, and explore important patterns within the dataset. The findings from EDA provide the foundation for the subsequent data preparation, model development, and evaluation stages of the project.




Distribution of Summary Length (Word Count)
Frequency
Number of Words in Summary
0
200
400
600
800
1000
1200
1400
1600
0
5
10
15
20
25
30


Review Length vs Star Rating
Review Word Count
Star Rating
0
20
40
60
80
100
120
1.0
2.0
3.0
4.0
5.0





3.3 Preparation steps Actually Performed

The dataset preparation was carried out before applying the sentiment classification models. The original Amazon product review dataset contained 4,915 records and 12 attributes, including reviewer information, product information, review text, rating, review time, and helpfulness-related attributes. The dataset was first loaded and its structure was examined to understand the available columns, data types, and number of records.

The first preparation step was to identify missing values in the dataset. A total of two missing values were found. One missing value was present in the reviewerName column and one missing value was present in the reviewText column. Since the review text is the main input required for sentiment analysis, the record with missing review text was removed. After this step, 4,914 usable reviews remained for further analysis.

The overall rating was then used to create the sentiment target variable. Ratings of 1, 2, and 3 were classified as Negative sentiment and assigned the label 0, while ratings of 4 and 5 were classified as Positive sentiment and assigned the label 1. This converted the original five-level rating information into a binary classification problem suitable for the machine learning and BERT models.

Additional text-related features were prepared for exploratory analysis. The review text was used to calculate review word counts and character counts, while the summary text was used to calculate summary word counts. Date-related information was also used to obtain the review year. These prepared features were used during EDA to examine review length, review frequency, and relationships between numerical attributes.

The prepared dataset was then divided into input features and target labels. The review text was used as the main textual input, while the newly created sentiment label was used as the target variable. The data was divided into training and testing sets using an 80:20 split with random_state=42 and stratification. This resulted in 3,931 training reviews and 983 testing reviews while maintaining the distribution of the sentiment classes.

For the traditional machine learning approach, the review text was converted into numerical features using TF-IDF. The TfidfVectorizer was configured with a maximum of 5,000 features, unigram and bigram features using ngram_range=(1,2), and English stop-word removal. These TF-IDF features were then supplied to Logistic Regression to establish a traditional NLP baseline.

For the BERT-based approach, the review text was prepared using the bert-base-uncased tokenizer. Reviews were tokenized with truncation and padding, with a maximum sequence length of 128 tokens. The resulting tokenized data was used as input to the BERT sequence-

classification model. The model was configured for two output classes corresponding to Negative and Positive sentiment.

Therefore, the preparation process transformed the original raw review dataset into a cleaned, labeled, analyzed, and model-ready dataset. The same prepared review data supported both the traditional TF-IDF and Logistic Regression baseline and the BERT-based sentiment classification approach.

3.4 EDA Exploration -Implementation

The Exploratory Data Analysis was implemented using Python libraries such as Pandas, Matplotlib, and Seaborn. The purpose of the implementation was to examine the structure and characteristics of the Amazon product review dataset and identify important patterns before applying the sentiment classification models. The analysis focused on rating distribution, review length, helpfulness information, review year, frequently occurring words, and relationships between numerical variables.

The first part of the EDA examined the distribution of review summary lengths. The number of words in each review summary was calculated and visualized using a histogram with a kernel density estimate. The resulting graph, titled “Distribution of Summary Length (Word Count),” helped show how the lengths of review summaries were distributed across the dataset. This provided an initial understanding of the amount of textual information contained in the summary field.

The relationship between review length and star rating was then explored using a boxplot. The graph titled “Review Length vs Star Rating” compared the number of words in reviews for each rating from 1 to 5. This analysis helped identify differences in review length across rating categories. The EDA showed that lower-rated reviews tended to contain longer reviews on average, while higher-rated reviews, particularly five-star reviews, were generally shorter.

The distribution of helpfulness-related information was also analyzed. Separate histograms were generated for total_vote and helpful_yes. These graphs showed how helpfulness votes

were distributed among the reviews and helped identify whether most reviews received only a small number of votes or whether some reviews had substantially higher vote counts.

The number of reviews across different years was examined using a count plot titled “Number of Reviews per Year.” The review_year feature was obtained from the review date information and used to determine how the number of reviews changed across different years. This provided an overview of the temporal distribution of the collected reviews.

Text-frequency analysis was performed to understand the words that occurred most frequently in the review text. A bar chart showing the “Top 25 Most Frequent Words in Review Text (all ratings combined)” was generated. This visualization provided an overview of commonly occurring terms in the complete review dataset and helped identify the general vocabulary used by customers.

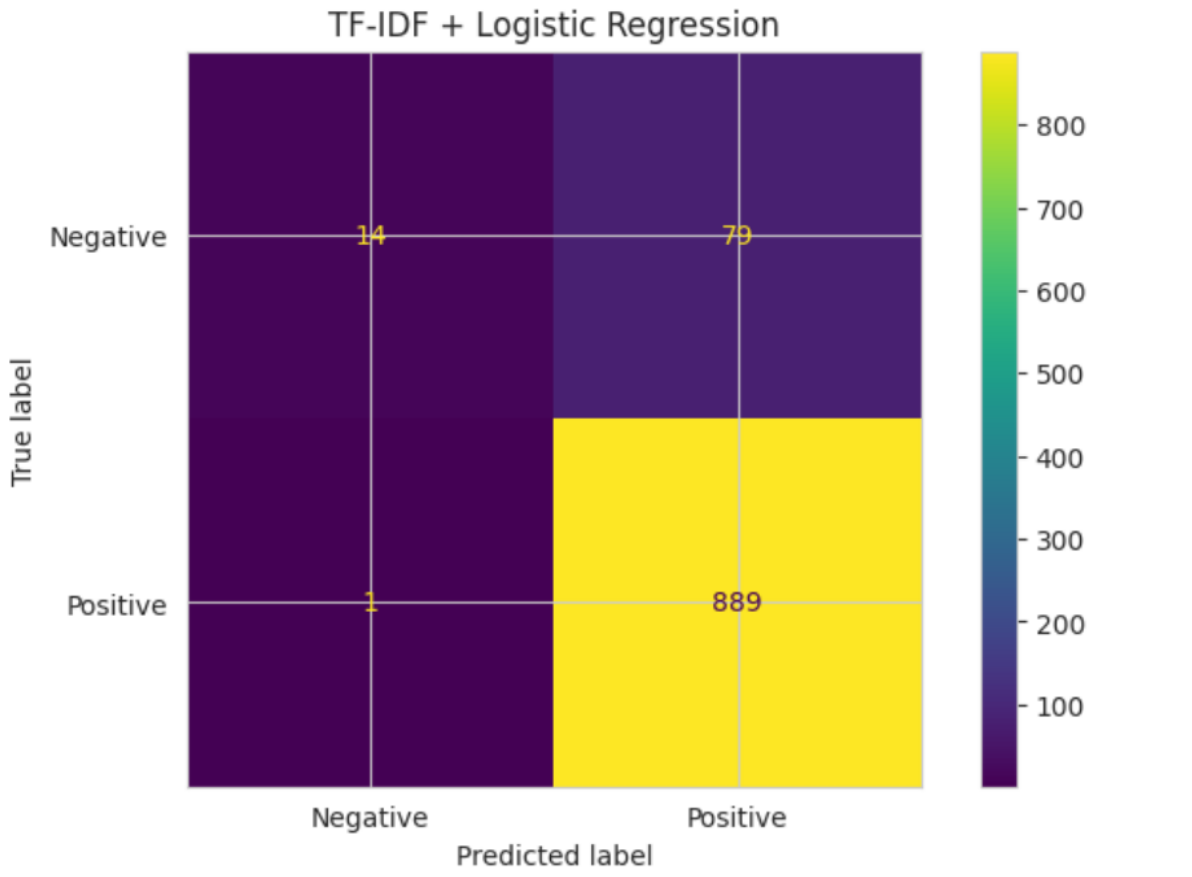
The frequent-word analysis was further separated according to sentiment-related rating groups. Reviews with ratings 4 and 5 were analyzed separately from reviews with ratings 1 and 2. Two bar charts were generated, titled “Top Words in Reviews with Rating 4-5” and “Top Words in Reviews with Rating 1-2.” Comparing these groups helped identify differences in frequently used vocabulary between higher-rated and lower-rated reviews.

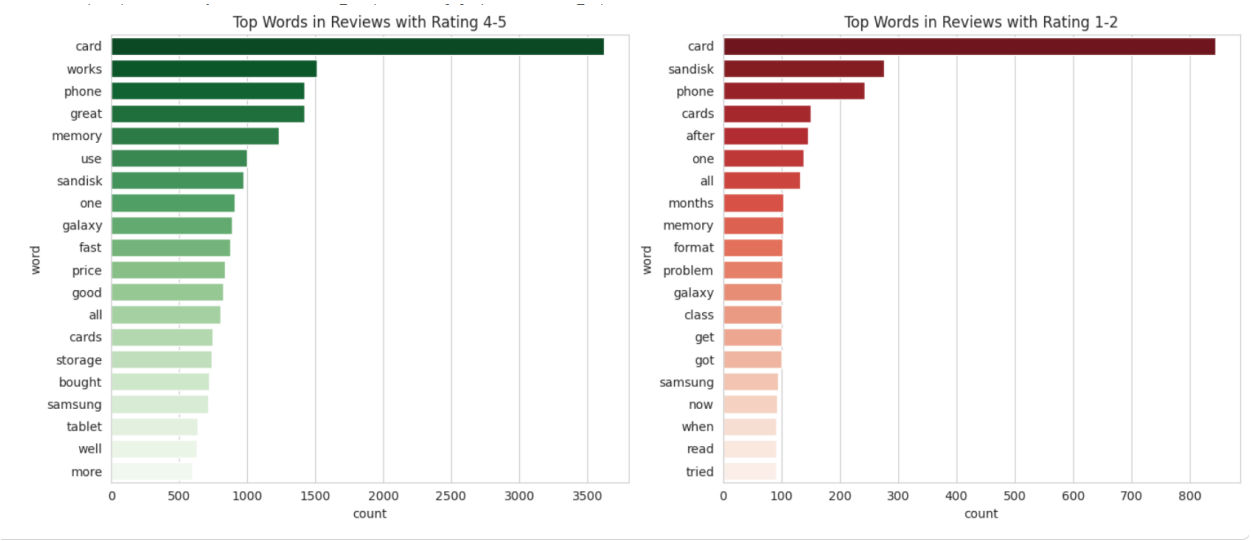
A correlation heatmap was also implemented for the numerical variables in the dataset. The graph was titled “Correlation Heatmap of Numeric Columns.” The analysis included variables such as overall, day_diff, helpful_yes, total_vote, review_word_count, review_char_count, and summary_word_count. The heatmap helped visualize the strength and direction of relationships between these numerical attributes and provided a compact view of the numerical structure of the dataset.

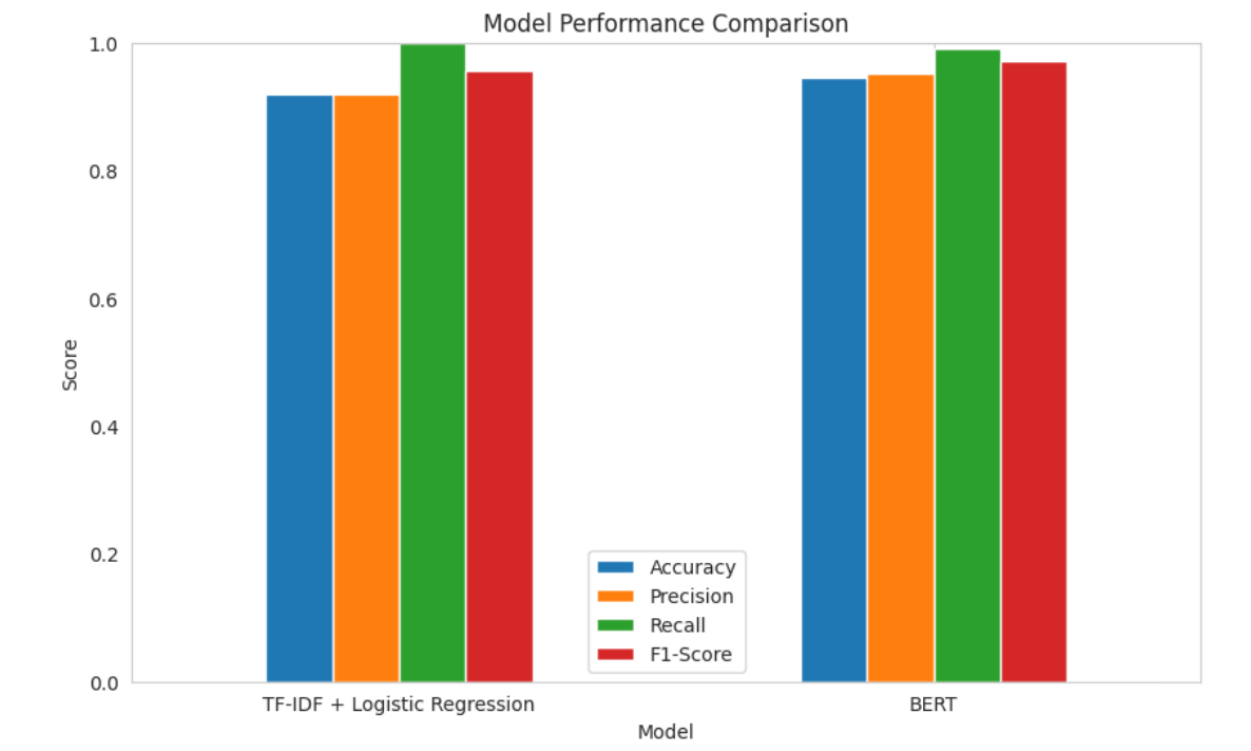
The EDA also examined the rating distribution and the resulting sentiment distribution. The analysis showed that five-star reviews were dominant in the dataset, with 3,922 reviews, while lower ratings occurred much less frequently. After converting the ratings into binary sentiment classes, there were 4,448 positive reviews and 466 negative reviews among the 4,914 usable reviews. This revealed a significant class imbalance, which was an important finding because

it could affect the performance of the sentiment classification models, particularly for the minority negative class.

Overall, the implemented EDA provided both numerical and visual understanding of the dataset before model development. The histograms, boxplots, count plots, word-frequency charts, and correlation heatmap helped examine text characteristics, rating patterns, helpfulness information, temporal distribution, vocabulary, and numerical relationships. These findings supported the subsequent preparation of the data and the implementation of both the TF-IDF with Logistic Regression baseline and the BERT-based sentiment classification model.







3.5 Summary of EDA Findings

The Exploratory Data Analysis provided several important findings about the Amazon product review dataset. The original dataset contained 4,915 reviews, and after removing the record

with missing review text, 4,914 usable reviews were available for analysis. The EDA helped understand the distribution of ratings, review length, helpfulness information, review frequency, word usage, and relationships between numerical variables before applying the sentiment classification models.

One of the most important findings was that five-star reviews were highly dominant in the dataset. There were 3,922 five-star reviews, representing approximately 79.80% of the usable reviews. The remaining ratings were distributed across one-star, two-star, three-star, and four-star reviews. This showed that the dataset was not evenly distributed across rating categories.

After converting the original ratings into binary sentiment classes, the dataset contained 4,448 positive reviews and 466 negative reviews. Therefore, positive reviews formed the majority class, while negative reviews represented the minority class. This class imbalance was an important EDA finding because a model could achieve high overall accuracy while still performing poorly when identifying negative reviews.

The analysis of review length showed a difference between rating groups. The average review word counts were approximately 103.86 words for one-star reviews, 114.48 words for two-star reviews, 73.25 words for three-star reviews, 56.21 words for four-star reviews, and 44.21 words for five-star reviews. This indicated that lower-rated reviews were generally longer, whereas highly rated reviews, especially five-star reviews, tended to be shorter.

The helpfulness analysis showed that the number of helpful votes and total votes varied across reviews. One-star reviews had a relatively high average number of helpful votes and total votes compared with several other rating groups. The histograms also showed that the vote distributions were uneven, with most reviews receiving relatively few votes while some reviews received substantially more votes.

The review-year analysis showed how the available reviews were distributed over time. This helped provide an understanding of the temporal coverage of the dataset and confirmed that the reviews were collected across multiple years rather than representing a single time period. The word-frequency analysis provided an understanding of the vocabulary used in the reviews.

The analysis of the top 25 most frequent words showed the most commonly occurring terms in the complete review collection. Separate analyses of reviews with ratings 4–5 and ratings 1–2 also helped compare vocabulary used in higher-rated and lower-rated reviews. These differences demonstrated that textual content contains useful information for distinguishing sentiment. The correlation heatmap provided an overview of relationships among numerical variables such as rating, review timing, helpful votes, total votes, review word count, review character count, and summary word count. This helped identify relationships between different numerical characteristics and provided additional understanding of the structure of the dataset.

Overall, the EDA showed that the dataset contained useful textual information for sentiment classification but also had a significant imbalance between positive and negative reviews. The strong dominance of five-star reviews, differences in review length across ratings, variation in helpfulness information, and differences in vocabulary between rating groups were important findings. These observations supported the decision to use the review text as the primary input for sentiment classification and to evaluate the models using multiple performance measures rather than relying only on accuracy. The findings also provided a clear foundation for comparing the traditional TF-IDF with Logistic Regression approach against the BERT-based sentiment classification model.

ALGORITHM SELECTION AND IMPLEMENTATION

The algorithm selection for this project was based on the objective of automatically classifying Amazon product reviews into positive and negative sentiment categories. Since the input data consists primarily of natural language text, Natural Language Processing techniques were required to convert the textual reviews into a form that machine learning models could process. Two approaches were implemented and compared: a traditional TF-IDF with Logistic Regression approach and a BERT-based sentiment classification approach. The traditional approach was used as a baseline, while BERT was selected to investigate whether a pretrained transformer model could provide better performance by understanding the context of review text.

The first approach implemented was TF-IDF with Logistic Regression. TF-IDF, or Term Frequency-Inverse Document Frequency, converts text into numerical features based on the

importance of words within individual reviews and across the complete collection of reviews. The implementation used TfidfVectorizer with a maximum of 5,000 features, unigram and bigram features using ngram_range=(1,2), and English stop-word removal. This representation allowed the textual reviews to be converted into numerical vectors suitable for machine learning.

Logistic Regression was then selected as the classification algorithm for the TF-IDF representation. It is a suitable baseline for binary text classification because it can learn the relationship between the TF-IDF features and the two sentiment classes. The model was configured with max_iter=1000 and random_state=42. The training data consisted of 3,931 reviews, while 983 reviews were reserved for testing. The model was trained using the TF-IDF features extracted from the training reviews and then evaluated on the unseen test reviews.

The second approach used BERT, which stands for Bidirectional Encoder Representations from Transformers. BERT was selected because traditional TF-IDF represents words primarily through their frequency and does not fully capture the contextual meaning of words. BERT uses a transformer-based architecture to process text while considering the surrounding words, making it more suitable for understanding contextual information in natural language reviews. The pretrained bert-base-uncased model was used for the implementation.
For the BERT implementation, the review text was first processed using the corresponding BERT tokenizer. Each review was tokenized with truncation and padding, with a maximum sequence length of 128 tokens. The pretrained BERT model was then configured for sequence classification with two output labels representing Negative and Positive sentiment. The model was fine-tuned on the prepared Amazon review dataset rather than being used only as a fixed feature extractor.

The BERT training configuration used a learning rate of 2e-5, a training batch size of 8, an evaluation batch size of 8, two training epochs, and a weight decay value of 0.01. The model was trained using the labeled review dataset and evaluated using the same held-out test set used for comparison with the traditional approach. This provided a consistent basis for evaluating the two methods.

The performance of both approaches was evaluated using accuracy, precision, recall, and F1-score. These metrics were considered together because the EDA showed a substantial imbalance between positive and negative reviews. Accuracy alone could therefore provide an incomplete picture of model performance, particularly for the minority negative class. Precision measures the correctness of positive predictions, recall measures the ability to identify relevant instances, and F1-score provides a balance between precision and recall.

The TF-IDF with Logistic Regression model provided a strong traditional baseline for the task. The BERT model achieved better overall performance in the implemented experiment, demonstrating the advantage of using contextual representations for product review sentiment classification. Based on the recorded evaluation results, the TF-IDF approach achieved approximately 91.86% accuracy and a 95.69% F1-score, while the BERT approach achieved approximately 94.51% accuracy and a 97.03% F1-score. Thus, BERT improved the accuracy by approximately 2.65 percentage points and the F1-score by approximately 1.34 percentage points compared with the baseline.

The project also implemented a prediction function for classifying new reviews. A new review is tokenized using the same BERT tokenizer and passed to the trained BERT model. The model produces output logits, which are converted into probabilities using Softmax. The class with the higher probability is selected as the predicted sentiment, and the corresponding confidence value is returned. This demonstrates how the trained model can be used beyond the fixed test dataset to classify newly provided product reviews.

Finally, the trained BERT model and tokenizer were saved using the save_pretrained() functionality so that the trained model could be reused without repeating the complete training process. Overall, the algorithm implementation followed an end-to-end workflow consisting of data preparation, TF-IDF baseline development, BERT fine-tuning, model evaluation, comparison of results, and prediction of new reviews. The comparison demonstrated that both approaches can perform sentiment classification, while the BERT-based approach provided better performance in the implemented experiment.

CONCLUSION

This case study successfully developed a product review sentiment analysis system using Natural Language Processing and BERT. The study used an Amazon product review dataset containing 4,915 original records. After handling the missing review text, 4,914 usable reviews were considered for analysis. The original five-level rating information was converted into two sentiment categories, where ratings from 1 to 3 were treated as Negative and ratings from 4 to 5 were treated as Positive.

The Exploratory Data Analysis provided important insights into the dataset. The analysis showed that five-star reviews were highly dominant, resulting in a significant imbalance between positive and negative reviews. Differences in review length were also observed across rating categories, with lower-rated reviews generally containing more words than higher-rated reviews. The analysis of helpfulness information, review years, frequently occurring words, and numerical relationships provided additional understanding of the characteristics of the dataset. These findings helped guide the preparation of the data and the selection of appropriate modelling approaches.

Two sentiment classification approaches were implemented and compared. The first approach used TF-IDF to convert review text into numerical features followed by Logistic Regression for classification. This provided a traditional NLP baseline and established a reference point for evaluating the second approach. The second approach used the pretrained bert-base-uncased model, which was fine-tuned for binary sentiment classification. BERT was selected because its contextual representation allows it to capture relationships between words more effectively than approaches based primarily on word frequency.

The models were evaluated using accuracy, precision, recall, and F1-score. In the implemented experiment, the TF-IDF with Logistic Regression approach achieved approximately 91.86% accuracy and a 95.69% F1-score, while the BERT approach achieved approximately 94.51% accuracy and a 97.03% F1-score. The results therefore showed that BERT provided better overall performance than the traditional baseline for the implemented dataset and experimental setup. However, the class-wise results also showed that identifying the minority negative class remained more challenging than identifying the majority positive class.

The project also demonstrated the practical use of the trained BERT model through a new-review prediction function. A new review can be tokenized, processed by the trained model, and classified as positive or negative using the predicted probability. The trained model and tokenizer were also saved so that they could be reused for future predictions without retraining from the beginning.

Overall, the study demonstrates an end-to-end sentiment analysis workflow, beginning with dataset preparation and EDA, followed by traditional NLP modelling, BERT fine-tuning, performance evaluation, model comparison, and prediction of new reviews. The results indicate that contextual transformer-based models such as BERT can provide improved sentiment classification performance for product reviews. At the same time, the limitations identified in the study, particularly class imbalance, rating-based sentiment labels, review ambiguity, and the maximum sequence length of 128 tokens, should be considered when applying the system to other datasets or real-world situations. Future work could focus on improving minority-class performance, using larger and more balanced datasets, exploring additional preprocessing and model configurations, and extending the system beyond binary sentiment classification.